

Assignment 1 – Draw lines [Part 2]

Description

For this project, simply unzip the contents of this zip file into the project directory. This will automatically replace the files that you have already created. Here are the changes:

- Added new GameStates.h
- Added SimpleLines.cpp, which replaces Level 1 from part 1 of this assignment.
- Added FreeDraw.cpp which allows to draw any line with the mouse.
- Added Common.h and Common.cpp containing GUI functionality.
- Modified Main.cpp to include the new game states.

What to do:

Modified DrawLine.h to include new functions for part 2

- `void DrawLine(const AVec2& p1, const AVec2& p2, const Color& c);`
- `void DrawLineNaive(const AVec2& p1, const AVec2& p2, const Color& c);`
- `void DrawLineDDA(const AVec2& p1, const AVec2& p2, const Color& c);`

You will have to implement the three methods. DrawLine should call the correspondent line scan conversion function depending on the method selected (DDA or Naïve) and the type of line (horizontal, vertical, diagonal or general).

Part of your task will be to make sure that the code is as optimized as possible, have a proper documentation with commentaries and use good code (see Code guidelines.pdf).

Submission Guidelines

Submit only DrawLine.cpp with a Readme.txt file as a zip folder on Moodle with the following naming conventions:

cs200fa25_{username}_1.1.zip

Additional guidelines

You have available in moodle the following files:

- "Code guidelines.pdf": a complete file detailing different aspect of the code etiquette I expect you to follow.

- "FileHeaderSample.h": example of a complete and correct header for two function and a file.
- "README_template.txt": template of the readme file you have to include in your submission.